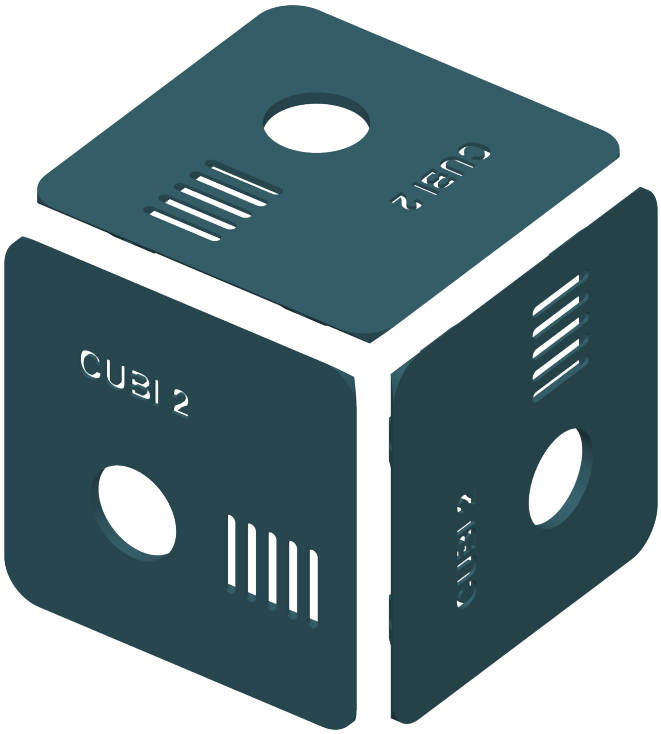
Materials

Everything to buy and to print for one Cubi 2



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All data come from tests and many real builds, not only from theory.

Teacher only

- **One thing per page**
  One action, one big picture. The part of the step is the only strong colour, the rest is quiet grey. Short sentences, much white space.
- **Colour is never the only clue**
  Every wire on a board drawing also has a number, every board a letter (X, Y, Z), every insert its size. The insert colours stay distinct with colour-vision deficiency (Okabe-Ito palette).
- **Readable**
  All instruction text has a contrast of at least 4.5 : 1 to the page (WCAG 2.x, AA).
- **Print it right**
  In colour, at 100 % - not 'fit to page'. The Assembly guide draws every screw at real size: the student lays the screw on the page to check it.
- **Pace**
  One page, then the action, then the next page. For a student who tires quickly: one chapter per session. The check pages are good stopping points.

In this list

For ONE Cubi 2. Each chapter: what it is, how many, where it goes.

- 1 Components - motors, drivers, battery, fuse, magnets
- 2 Electronics - the parts of the 3 boards (X, Y, Z)
- 3 Cables and small parts - every wire with its cut length, heat-shrink, solder
- 4 Fasteners - what to buy, then where each screw goes
- 5 Printed parts - every part, its colour and its settings

- All terminal blocks are 5.08 mm, with a clamp plate: no ferrules. Stranded ends twisted, no solder on them.
- Cut lengths are the route + 10 mm. Buy a little more of every wire.
- Screws: DIN 912 = socket head cap screw, DIN 7991 = countersunk, DIN 914 = grub screw with a cone point. Buy a few spares of the small ones.
- The printed parts are in CAD/print - the Printing guide shows each one on the bed.

1 · Components

Bought ready-made.

ITEM	SPECIFICATION	QTY	WHERE
BLDC motor	4006, 350 KV; bell Ø44.4 x 19.3; shaft Ø4; base 4x M3 on PCD 25, wheel 4x M3 on PCD 12	3	one per face, drives a reaction wheel
Motor controller	motor driver, integrated angle sensor encoder; mounting 4x M2.5 on 35.0 x 32.0	3	coaxial with each motor
Encoder magnet	diametric, Ø6.0 x 2.5	3	in the magnet carrier, on the motor shaft
IMU	6DoF IMU I2C; 25.40 x 25.40, 4x Ø3.2 on 20.32; I2C only, no header	1	on the battery tube base, on the main diagonal
Battery	6S 1400 mAh 150C, 22.2 V, battery connector; 76.0 x 39.0 x 40.5 (measured); 232 g	1	in the battery basin, on the main diagonal, hook-and-loop straps
Fuse	MIDI 30 A / 32 VDC; body 40 x 12 x 8; 2x M5 on 30	1	face x, main battery line
Ring terminal, insulated	DIN 46237, M5, 1.5 - 2.5 mm^2, blue; crimped	2	on the MIDI fuse, one per M5 screw
Battery connector pair	high-current connector, male + female, solder cups	1	male on the board-side wires (face z), female on the battery's own leads - or buy the battery with it already on
Hook-and-loop strap	20 x 250 mm	2	holds the battery in the basin, through the 4 slots in its back
Neodymium magnet	Ø3 x 2, N52	48	box covers: 24 in the covers, 12 in the box, 12 in the tie beams - CHECK POLARITY, every pair attracts
CA glue	cyanoacrylate	1	magnets into their Ø3.1 pockets

2 · Electronics

The parts of the 3 boards, as the Boards guide builds them.

ITEM	SPECIFICATION	QTY	BOARD
Perfboard	70 x 30 x 1.6, double-sided, 24 x 10 holes 2.54 pitch, 4x Ø2.2 corner holes	3	boards X, Y, Z - one per face
Terminal block 5.08 mm, 3-pole	PCB, 5.08 mm pitch, 3 poles, 16 A	7	Z: 2 (motors M1-M3) + 1 (out: SENSE, GND, +5V) · Y: 3 · X: 1 (CANL, CANH, CRX2)
Terminal block 5.08 mm, 2-pole	PCB, 5.08 mm pitch, 2 poles, 16 A	5	Z: 1 (battery) · Y: 1 · X: 3 (+5V, GND · BRX, BTX · CTX2, GND CAN)
DC-DC converter	5-36 V in, 5 V / 5 A out	1	Z: 22.2 V -> 5 V, turned over (marking up)
Pin header, female	1 x 6, 2.54 mm pitch, 8.5 mm high	1	Z: holds the DC-DC (its pins in row W)
MCU board	on 2 pin headers, 24 pins each	1	Y
Wi-Fi + Bluetooth MCU board	25 x 20.5 mm, on 2 x 8 pins	1	X, turned
CAN transceiver board	3.3 V, on 6 pins	1	X
TVS diode	1.5KE33CA, bidirectional 33 V 1500 W, DO-201	1	Z: protects +22V
Electrolytic capacitor	220 µF / 50 V, Ø10, polarized (long leg +)	1	Z: +22V
Fuse, axial	2 A, green body 8 x Ø2.2, soldered	1	Z: input of the DC-DC
Resistor, metal film 1 %, 1/4 W	100 kOhm + 10 kOhm, one each	2	Z: voltage divider for SENSE
Bare tinned copper wire	solid, 0.6 mm (22 AWG)	680 mm	X 270 · Y 160 · Z 250 mm (runs + 15 %)

2 · Electronics (continued)

ITEM	SPECIFICATION	QTY	BOARD
Bare copper, bus bars	1.5 mm², stripped silicone wire, filled with solder	110 mm	Z: GND and +22V bars
Insulated wire, thin	24 AWG, red	90 mm	Y: +5V on the top side, under the MCU

3 · Cables · to buy

The total of each wire, rounded up. Cut lengths are the route + 10 mm. Buy a little more of every wire.

WIRE	COLOUR	LENGTH
Silicone wire 0.75 mm²	black	700 mm
Silicone wire 0.75 mm²	red	800 mm
Silicone wire 0.75 mm²	white	400 mm
Silicone wire 1.5 mm²	black	600 mm
Silicone wire 1.5 mm²	red	500 mm
Silicone wire 14 AWG	black	100 mm
Silicone wire 14 AWG	red	400 mm
Solid wire 22 AWG	black	300 mm
Solid wire 22 AWG	blue	300 mm
Solid wire 22 AWG	green	300 mm
Solid wire 22 AWG	orange	300 mm
Solid wire 22 AWG	yellow	300 mm

3 · Cut list

Every wire, in the order of the Wiring guide: motor power, battery supply, board power, links.

WIRE	SPECIFICATION	CUT	FROM -> TO
M1+	1.5 mm² silicone, red	105 mm	driver X -> board Z, z1
M1-	1.5 mm² silicone, black	110 mm	driver X -> board Z, z2
M2+	1.5 mm² silicone, red	180 mm	driver Z -> board Z, z3
M2-	1.5 mm² silicone, black	200 mm	driver Z -> board Z, z4
M3+	1.5 mm² silicone, red	200 mm	driver Y -> board Z, z5
M3-	1.5 mm² silicone, black	220 mm	driver Y -> board Z, z6
Battery +	14 AWG silicone (Ø3.0), red	190 mm	z7 -> fuse
Battery +	14 AWG silicone (Ø3.0), red	210 mm	fuse -> battery connector
Battery -	14 AWG silicone (Ø3.0), black	100 mm	z8 -> battery connector
SENSE -> Y	0.75 mm² silicone (Ø2.1), white	350 mm	board Z SENSE -> board Y SENSE
GND -> Y	0.75 mm² silicone (Ø2.1), black	365 mm	board Z GND -> board Y GND
+5V -> Y	0.75 mm² silicone (Ø2.1), red	385 mm	board Z +5V -> board Y +5V
+5V -> X	0.75 mm² silicone (Ø2.1), red	345 mm	board Z +5V -> board X +5V
GND -> X	0.75 mm² silicone (Ø2.1), black	330 mm	board Z GND -> board X GND
Link BTX	22 AWG solid wire, blue	220 mm	board Y BTX -> board X BTX
Link BRX	22 AWG solid wire, orange	235 mm	board Y BRX -> board X BRX
Link CTX2	22 AWG solid wire, green	275 mm	board Y CTX2 -> board X CTX2
Link CRX2	22 AWG solid wire, yellow	265 mm	board Y CRX2 -> board X CRX2

3 · Ready-made cables

CABLE	SPECIFICATION	LENGTH	FROM -> TO
CAN 1	ready-made 4-wire cable, 1.25 mm plug	at least 60 mm	driver X (second plug) -> board X
CAN 2	ready-made 4-wire cable, 1.25 mm plug	at least 280 mm	driver X (first plug) -> driver Z
CAN 3	ready-made 4-wire cable, 1.25 mm plug	at least 260 mm	driver Z -> driver Y
Gyro (I2C)	ready-made 4-wire cable, 1.0 mm plug	at least 200 mm	IMU -> board Y, cut where its wires part
CAN termination plug	ready-made 4-wire cable, 1.25 mm plug, cut 3 cm from the plug	1	port 13 of driver Z

3 · Small parts

PART	SPECIFICATION	QTY	WHERE
Solid wire 22 AWG, 30 mm	a tail for each thin wire	7	I2C (4) at board Y, CAN 1 (3) at board X
Heat-shrink tube, small	over each tail joint	7	I2C (4), CAN 1 (3)
Heat-shrink tube	for the battery connector cups	4	2 board side, 2 battery side
Heat-shrink tube, small + one black over all		3	CAN termination plug
Resistor 120 Ohm	1/4 W, brown red brown gold	1	CAN termination plug
Solder	tin-lead 60/40 or 63/37, 0.6 - 0.8 mm, flux core	1	all joints
Masking tape	a name flag on every wire	1	

4 · Fasteners · to buy

Screws, nuts and heat-set inserts - the totals.

SIZE	TYPE	QTY
DIN 912 M2x5	socket head cap screw	12
DIN 912 M2.5x5	socket head cap screw	12
DIN 912 M2.5x6	socket head cap screw	2
DIN 7991 M3x6	countersunk, hex socket	54
DIN 912 M3x6	socket head cap screw	4
DIN 7991 M3x8	countersunk, hex socket	28
DIN 912 M3x8	socket head cap screw	18
DIN 7991 M3x10	countersunk, hex socket	2
DIN 7991 M3x12	countersunk, hex socket	3
DIN 912 M3x12	socket head cap screw	12
DIN 914 M3x12	grub screw, 90° cone point	8
DIN 7991 M3x14	countersunk, hex socket	3
DIN 7991 M3x16	countersunk, hex socket	6
DIN 912 M3x20	socket head cap screw	2
DIN 912 M5x6	socket head cap screw	2
DIN 912 M6x16	socket head cap screw	48
DIN 985 M6	nyloc nut	48
Insert M3 L 4.0	Heat-set insert, SHORT	91
Insert M3 L 3.0	Heat-set insert, SHORT 3 mm	3
Insert M3 L 5.7	Heat-set insert, NORMAL	14
Insert M2.5 L 4.0	Heat-set insert	14
Insert M2 L 3.0	Heat-set insert	12
Insert M5 L 5.8	Heat-set insert	2

4 · Fasteners · where

Each screw, where it goes and what it screws into.

WHERE	SCREW	QTY	INTO
Reaction wheel: rim masses	DIN 912 M6x16	48	nyloc nut
Wheel to motor bell (PCD 12)	DIN 912 M3x12	12	motor bell thread
Motor to Cubi (PCD 25)	DIN 912 M3x8	12	motor base thread
Motor drivers to their standoffs	DIN 912 M2.5x5	12	M2.5 insert
Boards on their standoffs, faces z, x, y	DIN 912 M2x5	12	M2 insert
IMU board into the table (heads inside the pocket)	DIN 912 M3x6	4	M3 SHORT insert L4.0 (2.4 engaged)
Battery basin floor into the table	DIN 7991 M3x8	4	M3 SHORT insert L4.0 (4.0 engaged)
MIDI fuse terminals	DIN 912 M5x6	2	M5 insert
Hub standoff flange to the basin top plate	DIN 912 M3x8	3	M3 SHORT insert L3.0
Battery tube to corner 0, from outside	DIN 7991 M3x12	3	M3 SHORT insert
clamp_power bar to the box (face z)	DIN 912 M2.5x6	2	M2.5 insert
Diagonal strut flange to the corner 0 floor, from outside	DIN 912 M3x8	3	M3 SHORT insert L4.0
Cage: beams to corners	DIN 7991 M3x8	24	M3 SHORT insert
Cage to Cubi	DIN 7991 M3x14	3	M3 NORMAL insert
Corner tips - the cube alone, out again before the sphere	DIN 914 M3x12	8	M3 NORMAL insert (corner tip), 6.2 mm out
Sphere: arcs	DIN 7991 M3x6	48	M3 SHORT insert
Sphere: diagonal arc crossings	DIN 7991 M3x6	6	M3 SHORT insert in the lower arc
Sphere: navels, 6 corners	DIN 7991 M3x16	6	M3 NORMAL insert (corner tip)

4 · Fasteners · where (continued)

WHERE	SCREW	QTY	INTO
Sphere: navels, corners 0 / (1,1,1)	DIN 7991 M3x10	2	M3 NORMAL insert (corner tip)
Diagonal strut / hub standoff to the corner	DIN 912 M3x20	2	M3 SHORT 4 mm insert, inside the corner
Reaction wheel: rim masses	DIN 985 M6	48	on the M6x16

4 · Heat-set inserts · where

INSERT	SPECIFICATION	QTY	WHERE
Heat-set insert, SHORT M3	L 4.0, hole Ø4.00	91	cage, sphere (navels + lower diagonal arcs), table (IMU pocket + basin screws), corner 0 floor (diagonal strut flange), inside cage corners 0 and (1,1,1) (strut / hub standoff screw)
Heat-set insert, SHORT 3 mm M3	L 3.0, hole Ø4.00	3	basin top plate (hub standoff flange)
Heat-set insert, NORMAL M3	L 5.7, hole Ø4.00	14	cage-to-Cubi ties, cage corner tips, battery foot (3)
Heat-set insert M2.5	L 4.0	14	motor driver standoffs, cable clamp base
Heat-set insert M2	L 3.0, OD Ø3.5	12	board standoffs
Heat-set insert M5	L 5.8	2	fuse bosses

5 · Printed parts

79 parts

PETG-CF. Settings: layer · walls · infill. Grams: one piece (weighed, or from the model). The Printing guide has the details.

PART	FILE	QTY	COLOUR · SETTINGS · GRAMS
Box	box.step	1	Milk Tea Brown · 0.16 · 4 walls · 20% · 172.0 g weighed
Cage corner	corner.step	8	Ancient Wood Green · 0.08 · 5 walls · 20% · 2.6 g
Cage beam	beam.step	9	Milk Tea Brown · 0.16 · 4 walls · 20% · 9.2 g
Cage beam with Cubi tie hole + 2 cover magnet pockets	beam_tie.step	3	Milk Tea Brown · 0.16 · 4 walls · 20% · 9.0 g
Box cover, magnetic, inset	cover.step	3	Ancient Wood Green · 0.08 · 5 walls · 20% · 19.2 g
Reaction wheel	reaction_wheel.step	3	Ancient Wood Green · 0.16 · 4 walls · 90% · 94.0 g
Magnet carrier	magnet_carrier.step	3	Ancient Wood Green · 0.08 · 5 walls · 20% · 0.6 g
Sphere arc, edge	sphere_arc_edge.step	12	Milk Tea Brown · 0.16 · 4 walls · 20% · 7.5 g
Sphere arc, diagonal (under)	sphere_arc_diag_under.step	6	Milk Tea Brown · 0.16 · 4 walls · 20% · 12.9 g
Sphere arc, diagonal (over)	sphere_arc_diag_over.step	6	Milk Tea Brown · 0.16 · 4 walls · 20% · 12.8 g
Navel	navel.step	8	Ancient Wood Green · 0.08 · 5 walls · 20% · support · 8.2 g
Battery tube, lower support + IMU base (star foot)	battery_tube.step	1	Ancient Wood Green · 0.16 · 4 walls · 100% · 22.5 g
Battery basin (IMU screws, hub standoff) - PRINT ON ITS BACK	battery_basin.step	1	Milk Tea Brown · 0.16 · 4 walls · 100% · 29.7 g
Battery tray standoff, upper	battery_top_standoff.step	1	Ancient Wood Green · 0.08 · 5 walls · 20% · 3.3 g

5 · Printed parts (continued)

79 parts

PART	FILE	QTY	COLOUR · SETTINGS · GRAMS
Box standoff, corner 0	box_standoff.step	1	Ancient Wood Green · 0.08 · 5 walls · 20% · 5.8 g
clamp_power - cable clamp bar (battery leads)	clamp_power.step	1	Milk Tea Brown · 0.16 · 4 walls · 20% · 0.3 g
clamp_z - cable clamp	clamp_z.step	1	Ancient Wood Green · 0.16 · 4 walls · 20% · 2.0 g
clamp_y - cable clamp	clamp_y.step	1	Ancient Wood Green · 0.16 · 4 walls · 20% · 2.7 g
clamp_x_2 - cable clamp	clamp_x_2.step	1	Ancient Wood Green · 0.16 · 4 walls · 20% · 1.2 g
clamp_x_1 - cable clamp	clamp_x_1.step	1	Ancient Wood Green · 0.16 · 4 walls · 20% · 0.9 g
Clamp spacer OD 9.5 / ID 7 / H 2 (on each clamp ring, under the board)	spacer_OD9.5_ID7_H2.step	8	Ancient Wood Green · 0.16 · 4 walls · 20% · 0.1 g